

Usability testing findings

AI-driven clinical decision-making simulation for undergraduate nursing education

1. Summary

Three moderated sessions were run against the Figma prototype. The scenario itself worked: all three participants understood what was being asked and both nursing participants recognised it as outside student scope without prompting. The problems were in the surrounding interface rather than the scenario design. Five issues were logged; three are being fixed before the demo, one is specified for Dev, and one is backlogged. The most serious is that participants did not notice the exit control.

2. How the sessions ran

Three moderated think-aloud sessions of approximately 30 minutes. Participants worked through the prototype without guidance, followed by structured questions and a debrief.

Method, session structure and participant care followed the usability test plan. One deviation: P3 is not from the nursing cohort, so their session informs the interaction and comprehension questions only and cannot speak to whether the pressure feels realistic.

Participants

ID	Background	Placement experience	Path taken
P1	IT, year 3	No	Refused all three — NO outcome
P2	IT, year 3	No	Agreed at pushback 2 — YES outcome
P3	IT, year 3 (non-nursing)	No	Asked questions, then used the exit

Participants are identified by code only.

3. What the testing had to answer

The six questions from the test plan, and what the sessions showed against each.

Area	Question	What the sessions showed
Comprehension	Did the student understand what Sandra was asking, and that it fell outside their scope?	Held up. All three identified the request as IV administration;
Interaction	Did they know how to respond, and that it was free text?	Did not hold. Two of three waited for response options to appear before typing.
Pressure	Did the escalation feel real enough to be uncomfortable?	Held up. P1 rewrote their reply twice at pushback 3 and described the threat afterwards as "exactly how it sounds in real life".

Area	Question	What the sessions showed
Exit	Did they know they could leave, and would they feel able to?	Did not hold. Only P3 noticed the control, and then only when probed. P1 said they would have closed the tab. See F2.
Resolution	Did the outcome screens explain the decision rather than just rate it?	Partly. P2 scrolled past the model answer looking for a continue button and did not return to it. See F3.
Unclear input	Did the flow hold when the student neither agreed nor refused?	Inconclusive, and the prototype limited this. P3 asked two questions and found Sandra's fixed reply dismissive of them. See F4 and section 11.

The sixth question is the one that most needed evidence, and it is the one the prototype was least able to answer. See section 11.

4. Session summaries

Descriptive only — interpretation is in section 6.

P1

Path taken. Refused at the opening request and at all three pushbacks. Reached the NO outcome.

What they did. Read the entry screen in full, then typed a short refusal citing scope within about twenty seconds. Became slower at each pushback. At pushback 3 they typed, deleted and retyped their reply twice before sending.

Notable moments. Visibly uncomfortable at pushback 3 — leaned back from the screen before answering. When asked afterwards where they would go to stop, they said they would have closed the browser tab.

In their words. "That is exactly how it sounds in real life. The bit where she says she will tell your facilitator is the part that actually gets you."

P2

Path taken. Refused once, then agreed at pushback 2. Reached the YES outcome.

What they did. Paused at the ward screen for roughly fifteen seconds before typing, and asked the facilitator whether there were options to choose from. Refused once, hesitantly. After pushback 2 they agreed, saying they did not want to seem difficult.

Notable moments. On the outcome screen they scrolled straight to the bottom looking for a continue button, then stopped. When asked what the screen had said, they could describe the verdict but not the model answer.

In their words. "I knew it was probably wrong but she is the one in charge. I would not want to be the student who argues."

P3

Path taken. Asked two questions rather than answering, then used the exit when probed.

What they did. Asked what was in the bag, then asked whether someone else could do it. Sandra's scripted reply did not acknowledge either question, which P3 described as being talked over. Did not find the exit control until asked where they would go to stop.

Notable moments. As a non-nursing participant they could not judge whether the clinical framing was realistic, and their session is not counted toward the pressure question.

In their words. "She never actually answered me. In a real conversation I would have just kept asking until she did."

5. Probe results

Paths participants did not reach naturally, attempted as a second pass.

Probe	Who	What happened
Asked a question instead of answering	P3	Asked what was in the bag. Sandra's scripted line did not acknowledge the question, which P3 read as being stonewalled rather than pressured. See F4.
Agreed immediately	—	Not reached. P2 agreed at pushback 2 rather than at the opening request. Run this probe in the next round.
Refused all three times	P1	Escalation built as intended. P1 described pushback 3 as the point where they "started doubting themselves", which is the effect the design intends.
Refused then agreed	—	Not reached. Coerced-agreement screen untested.
Used the exit	P3	Found it only after being asked where they would go to stop. Once used, described the screen as "a relief" and noted it did not tell them off.

Two of five probes were not reached with three participants. With a larger round, assign probes deliberately rather than hoping they come up.

6. Findings

Log

ID	Screen	What happened	Affected	Severity
F1	Ward interface	Participants waited for response options instead of typing	P2, P3	Major
F2	Ward interface	Exit control not noticed; one participant would have closed the tab	P1, P2	Critical
F3	YES outcome	Model answer scrolled past while looking for a continue button	P2	Major
F4	Escalation	Sandra's fixed reply read as ignoring a direct question	P3	Major
F5	Entry screen	Ward time "14:02" read as a countdown by one participant	P3	Cosmetic

Severity was assigned after all sessions, not during.

F2 - Exit control not noticed

Severity: Critical · **Screen:** Ward interface · **Affected:** P1, P2

What happened. Neither nursing participant noticed the leave control. Asked afterwards what they would have done to stop, P1 said they would have closed the browser tab and P2 said they assumed stopping was not allowed.

Why it matters. This bears on the exit question, and it is the one finding with a wellbeing dimension rather than only a usability one. The design claims a student can always leave; if they cannot find the control, that claim does not hold.

Suspected cause. The control sits in the top bar chrome, deliberately de-emphasised so it does not read as an answer option. It was de-emphasised too far.

Proposed change. Increase prominence and use the full label "Leave scenario" outside the chrome. Re-test specifically for whether it now reads as a third answer option, which is the failure mode in the other direction.

F1 - Participants waited for response options

Severity: Major · **Screen:** Ward interface · **Affected:** P2, P3

What happened. Two participants paused at the ward screen without typing. P2 asked the facilitator whether there were options to choose from.

Why it matters. Bears on the interaction question. Free text is a core requirement, and a participant who waits for options loses the first and most honest reaction to the request.

Suspected cause. The entry screen says to answer in their own words, but it appears before the ward screen and is not reinforced once Sandra speaks.

Proposed change. Add a short cue in the reply field or immediately below it making clear the reply is typed. Avoid suggesting phrasings, which would defeat the objective.

F4 — A direct question read as stonewalling

Severity: Major · **Screen:** Escalation states · **Affected:** P3

What happened. P3 asked two direct questions. Sandra's fixed prototype reply did not acknowledge either, and P3 described the exchange as being talked over rather than pressured.

Why it matters. Bears on the unclear-input question — the one design decision made without client or requirements authority. The intended behaviour is that Sandra holds her position without escalating; what P3 experienced was closer to the scenario ignoring them.

Suspected cause. Cannot be separated from the prototype limitation. Sandra's replies are fixed here, so a question receives the same line as a refusal.

Proposed change. Not a prototype fix. Specify to Dev that Sandra acknowledges the question in character before restating her request, and re-test on the built version.

F3 and F5 are recorded in the log without a detail section. Expand a finding here only when the cause or the proposed change needs explaining.

7. What worked

Design decisions the sessions supported, recorded so they are not undone later.

- No participant looked for a progress indicator or asked how many times they would need to refuse. The decision to omit one held.
- The escalation read as pressure rather than as the system judging the answer. P1 described doubting themselves at pushback 3, which is the intended effect.
- The exit screen, once reached, was described as a relief and explicitly as not telling the participant off. The content is right even though the route to it is not.
- Both nursing participants identified the scope-of-practice issue unprompted, suggesting the entry briefing gives enough context without hinting at the answer.

8. Triage

ID	Finding	Decision	Change
F2	Exit control not noticed	Must fix before demo	Increase prominence and label it "Leave scenario" in full, outside the top bar chrome.
F1	Participants waited for options	Must fix before demo	Add a one-line cue on the entry screen that replies are typed in their own words.
F3	Model answer skipped	Must fix before demo	Move the model answer above the explanation block on the YES outcome.
F4	Question read as stonewalling	Flag to Dev, not a prototype fix	Prototype cannot test this properly. Specify that Sandra acknowledges the question before restating.
F5	Ward time misread	Backlog	Cosmetic, one participant, non-nursing. No change.

9. Changes made

1. Exit control made more prominent and given the full label, addressing F2. 
2. Typed-reply cue added to the ward screen, addressing F1.
3. Model answer moved above the explanation block on the YES outcome, addressing F3.

10. Limitations

- **Prototype, not a build.** Sandra's replies were fixed, so participants typed freely but received a scripted response. Findings about her wording hold; findings about her responsiveness do not. F4 sits on that line and should be re-tested once the AI is running.
- **Three participants.** Enough to surface obvious problems, not enough to quantify anything. No percentages are reported, and a finding from one participant is flagged as such.
- **One participant outside the cohort.** P3 is not a nursing student, so their session informs interaction and comprehension only. They cannot judge whether the clinical pressure is realistic.
- **Two probes unreached.** The coerced-agreement screen and immediate agreement were not tested at all.